

INTRODUCTION TO OCPP 1.6



GOAL

The goal of this course is to learn what the OCPP 1.6 protocol is, how it works, and what are key messages of the protocol.

The course focuses on the key messages of the protocol is, helping you to get started working with the protocol.



CONTENTS

1. Basic concepts and terminology
2. Connecting charging stations to a central system
3. Station's online-state and heartbeats
4. Station status management
5. Starting a new charge with an RFID tag
6. Starting a new charge with a mobile app
7. Getting energy readings and other data during charging
8. Stopping a charge with an RFID tag
9. Stopping a charge with a mobile app
10. Recap and future learnings



OCPP documentation



It is recommended that you **download the OCPP documentation** as a support material of this course.

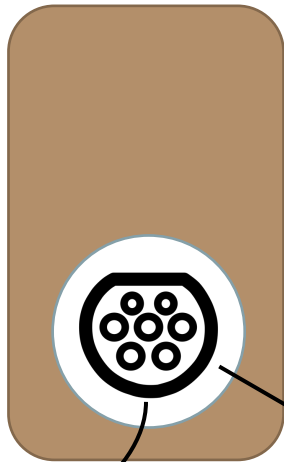
You can find the documentation at
<https://www.openchargealliance.org>
(Protocols -> Downloads -> OCPP 1.6)



1. BASIC CONCEPTS AND TERMINOLOGY



Charging Station and Connector



Electric Vehicle (EV)

A **charging station** is a physical device and system that allows electric vehicles (EVs) to be charged.

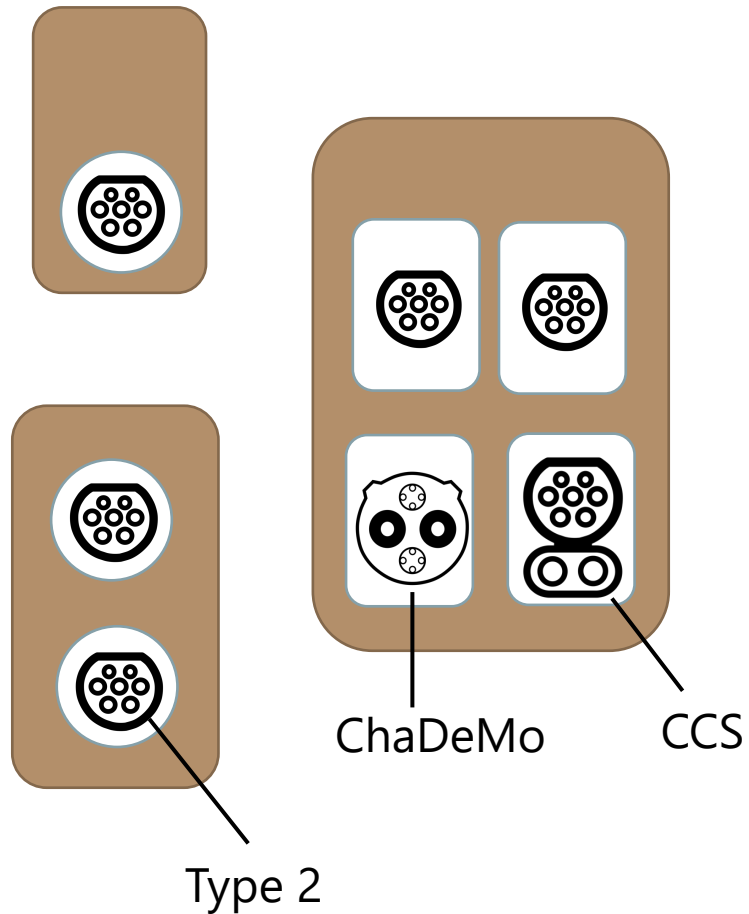
OCPP 1.6 uses the term 'charging point' instead of 'charging station'.

This term has been replaced by the term 'charging station' in newer OCPP versions.

A **charging station** may have one or more connectors. One electric vehicle (EV) can be plugged into one connector.



Different Types of Charging Stations



There are countless of different charging station **models** with **different configurations** from different **vendors**.

A charging station can have **one or more connectors**. There are several different connector types, such as CSS and ChaDeMo for fast DC charging, or Type 2 for slower AC charging.



Some Examples of How Charging Stations Look in Real Life



Picture by Ivan Radic
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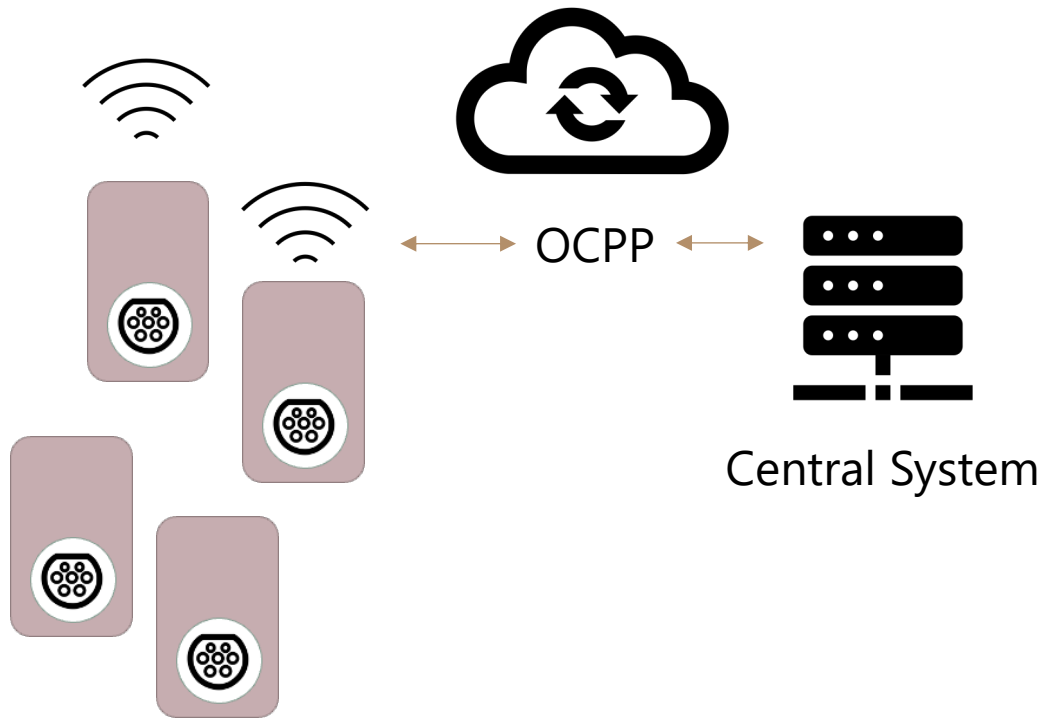
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Central System



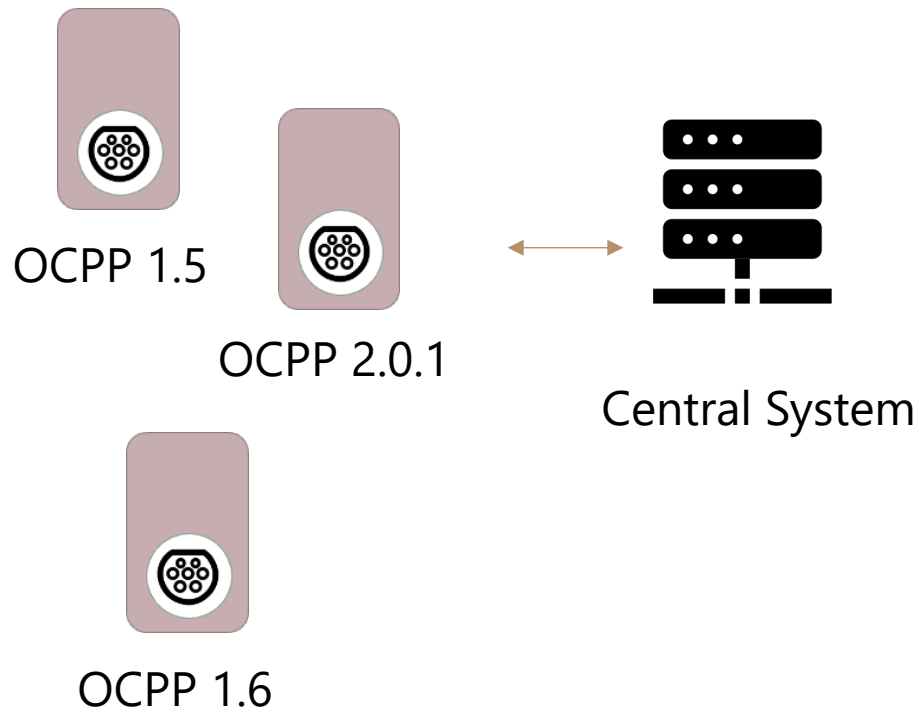
Charging stations may have an internet connection, which can be established through various methods such as fixed LAN connections, WiFi, or mobile networks.

A Central System is an IT system used to control multiple charging stations. Central systems perform various tasks, such as authorizing whether or not an EV driver is allowed to use a station.

The communication protocol between charging stations and a central system is called **OCPP**.



OCPP Versions



OCPP has **several versions** that are used by different charging stations. Older versions than 1.5 are rare, and version 1.5 is slowly being phased out of the market.

The most common version, used by hundreds of thousands of charging stations today, **is version 1.6**. This course also focuses on this version.

The new 2.x versions are slowly being implemented in the market, but progress has been slow. In 2023-2024, most companies are still expected to be operating systems with version 1.6.

